AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 33 · STUDENT LESSON

# Capstone Design

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

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## Before you start

Weeks 19 and 26: specify a need and difficult tests before building.

Plan time to read, investigate and explain. Keep predictions separate from observations.

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STUDENT LESSON · 1 OF 18

## Your mission

> Start with the target: Explain the mechanism and terminology of capstone requirements and corpus. Try the recall before opening notes.

### Your learning target

I can explain the **mechanism** and terminology of capstone **requirement**s and corpus.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Human AI Workflow. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 33: Capstone Design

**Requirement**s, **success measure**s, sources, risks, and tests precede building.



CREATOR MISSION

Complete a product brief and **design review**.

Creator name and date:

> **Write or draw your reasoning and evidence.**

### Readiness for the connected explanation

Recall System evaluation sets. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

### This week's end goal

By the end, I can explain capstone design using a traced example and a stated limitation.

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STUDENT LESSON · 2 OF 18

## Notice the evidence

> Investigate the supplied Capstone Design case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



**Start with this puzzle**

MYSTERY

The learners must use capstone design to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

**Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

**Write or draw your reasoning and evidence.**

### A closer explanation

A grounded-assistant capstone begins with bounded **requirement**s and a versioned permitted corpus. A baseline can be a manual lookup **process** using the same **evidence**.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

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STUDENT LESSON · 3 OF 18

## See how it works

> Follow the mechanism for Capstone Design. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

**Input → Process → Output**

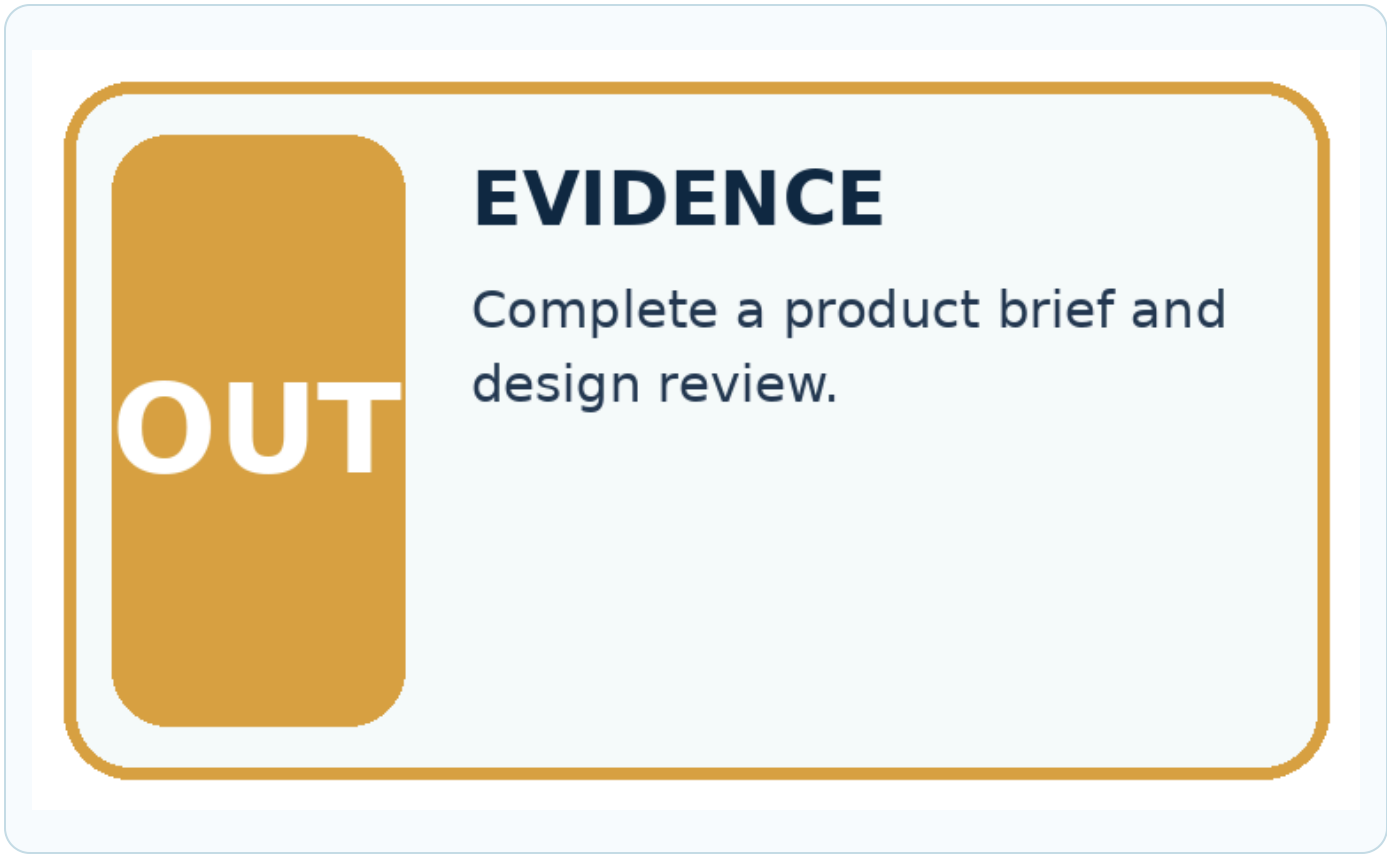
Trace the information. At each stage, ask what changed and how you could test it.




↓


SYSTEM
Requirements, success measures, sources, risks, and tests precede building.

↓


OUT
EVIDENCE
Complete a product brief and design review.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

### A closer explanation

Specify the user question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval.

### Capstone requirements and corpus — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review. | Specify the user question, source authority, access, output/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation. Keep teacher material out of student retrieval. | Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

### Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

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STUDENT LESSON · 4 OF 18

## Compare the cases

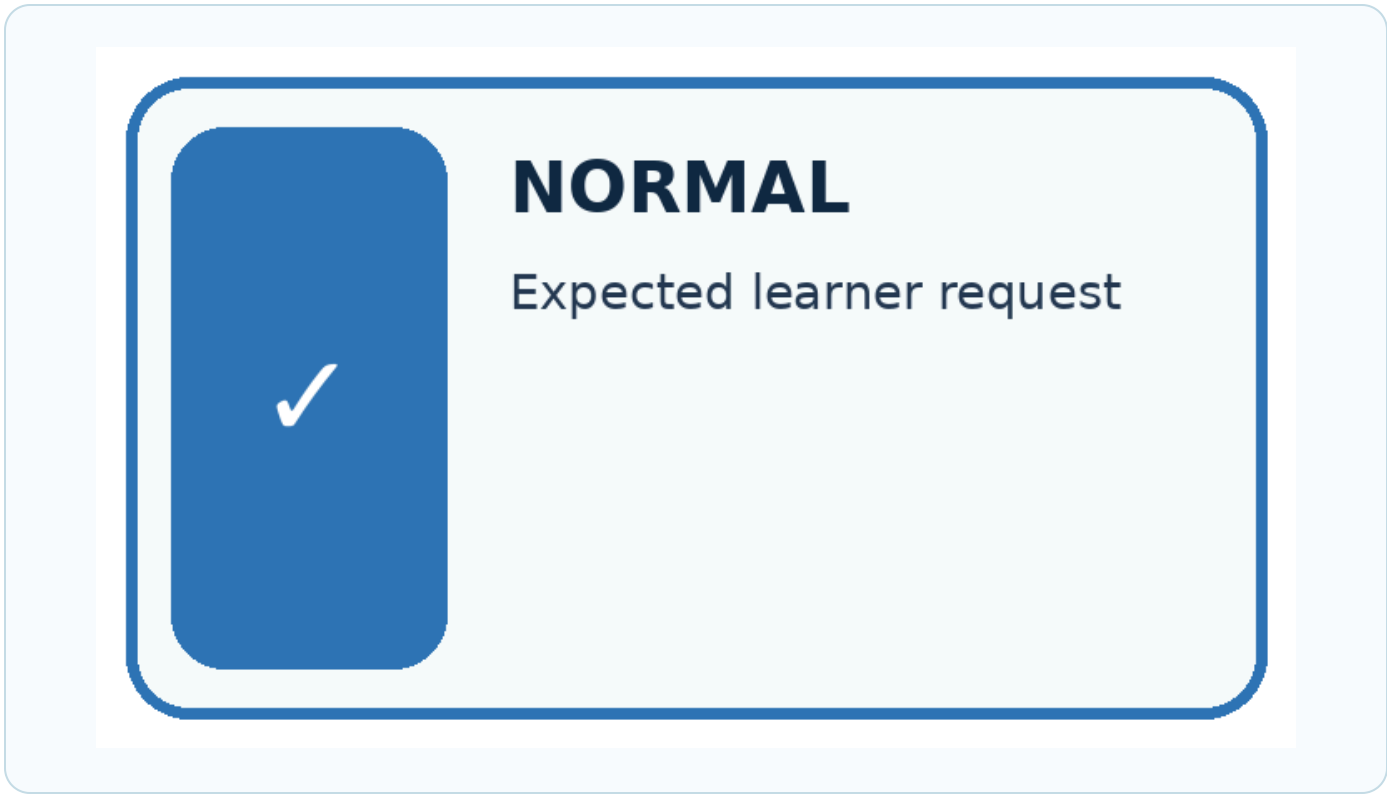
> Compare the supplied cases for Capstone Design. Identify the detail that changes the conclusion.

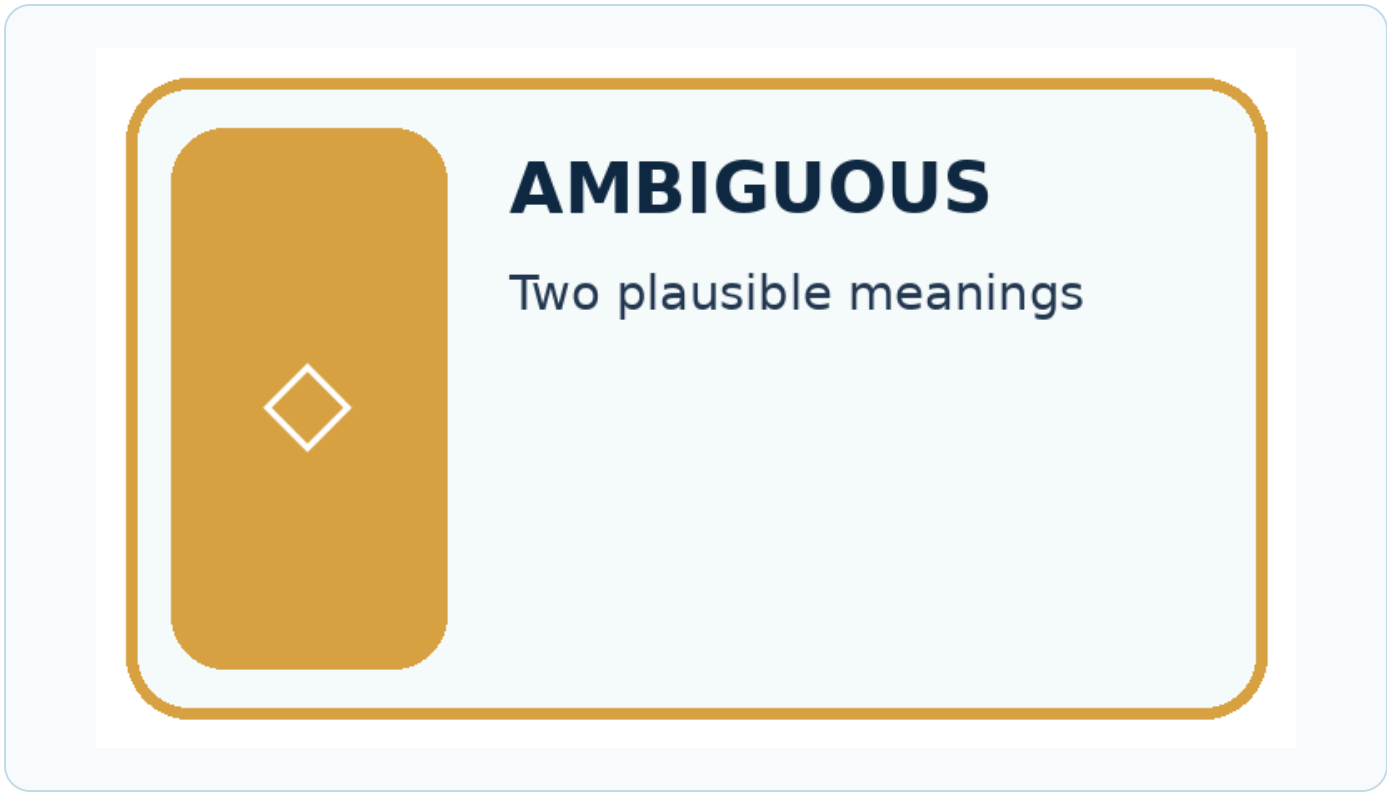
### The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.


NORMAL
Expected learner request


AMBIGUOUS
Two plausible meanings

### The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.






### Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

**Change one thing in the pictured case**

### Change one thing

Change: The corpus gains a teacher-approved flower definition in V2.

### Predicted result

**Prediction**: Add a flower test using V2 and record the corpus version.

### Why the result follows

Reason: The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

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STUDENT LESSON · 5 OF 18

## Words that unlock the idea

> Use the vocabulary of Capstone Design to explain a concrete decision, rather than only memorising definitions.

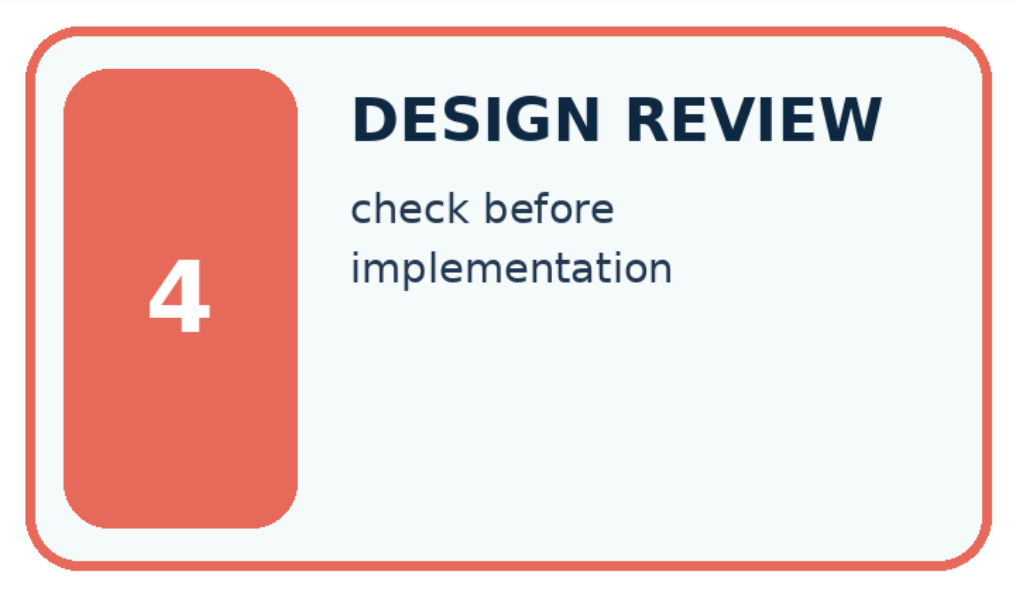
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.


1
REQUIREMENT
behavior the product must provide
2
SUCCESS MEASURE
observable evidence of value




| Word | My own example | Not this / boundary |
|---|---|---|
| requirement | A behaviour or condition the proposed system must satisfy. | |
| success measure | An observable quantity or criterion used to decide whether a task was achieved. | |

| Word | My own example | Not this / boundary |
|---|---|---|
| risk register | A record of identified risks, their consequences, mitigations and responsible reviewers. | |
| design review | An examination of a proposed system against its requirements before proceeding. | |

### Words used in the closer explanation

| Word | Meaning |
|---|---|
| Corpus manifest | An inventory of source IDs, versions and access roles |
| Baseline procedure | A credible reference process on the same task |

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STUDENT LESSON · 6 OF 18

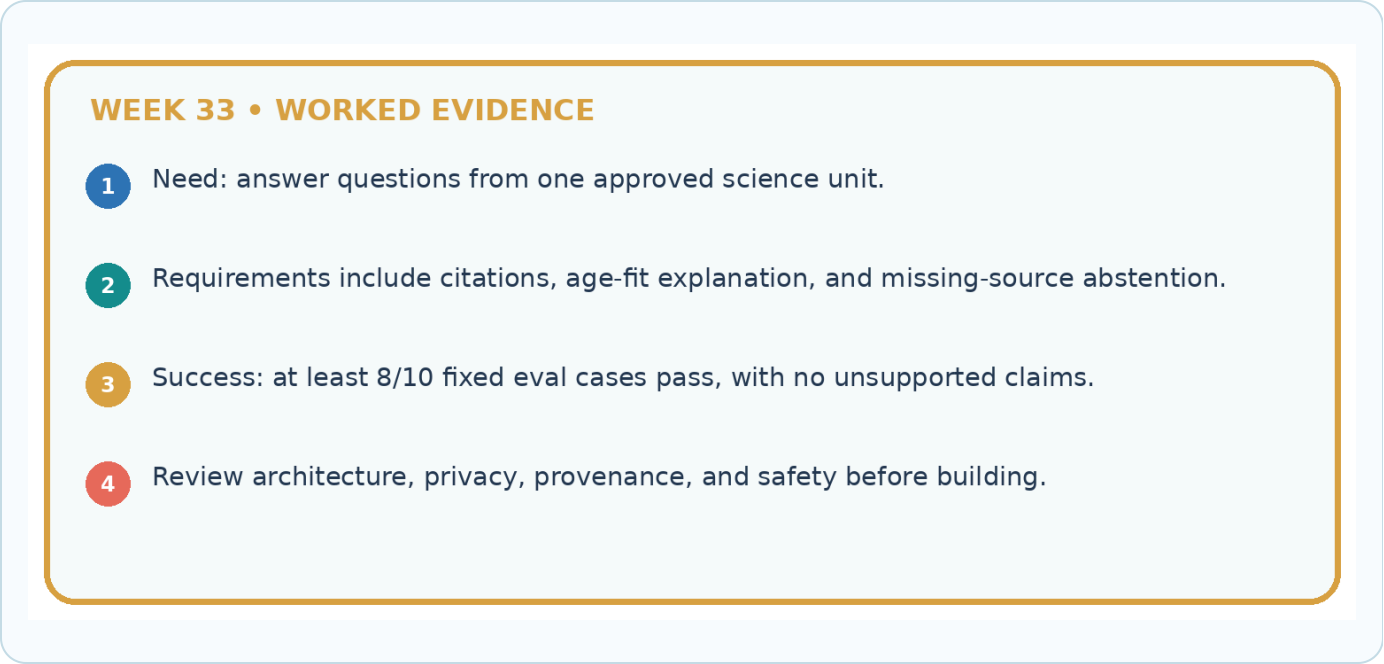
# Follow a worked example

> Follow the worked trace for Capstone Design, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.




1. Need: answer questions from one approved science unit.

2. **Requirement**s include citations, age-fit explanation, and missing-source abstention.

3. Success: at least 8/10 fixed eval cases pass, with no unsupported claims.

4. Review architecture, privacy, provenance, and safety before building.

Recalculate, retrace, or restate the reasoning in your own representation:

**Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

### A closer explanation

Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review.

Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity. Reserve complete, missing and conflicting cases before development.

### Worked explanation · Capstone Design

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Capstone Design

Plan a bounded study buddy around a small approved vocabulary corpus.

**The facts we will use**

### Supplied facts

Corpus V1 defines roots, stem and leaf. It contains no plant disease treatment. Success requires a source reference; missing definitions require a clear fallback.

### Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-3/33

# Capstone Design

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Define contract

Answer only glossary-definition requests supported by V1.



## 2. Plan normal test

"What is a stem?" must point to V1's stem entry.



## 3. Plan missing test

"How do I cure this disease?" must produce an out-of-scope response, not a treatment guess.

**CHANGE ONE THING**

The corpus gains a teacher-approved flower definition in V2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Define contract | Answer only glossary-definition requests supported by V1. |
| 2. Plan normal test | “What is a stem?” must point to V1’s stem entry. |
| 3. Plan missing test | “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### Why the result follows

Read the picture in order. “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

**What this example does not establish**

### Keep the stated limit

A capstone design is a testable plan; its proposed behaviours are not verified outcomes until checked.

### Keep the stated limit

Read the labelled picture: What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

### Keep the stated limit

Change one thing: The corpus gains a teacher-approved flower definition in V2. Predict the result before checking the explanation. What remains the same?

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STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the Capstone Design claim and explain a repair using the supplied evidence.

### Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

## COMPARE • DIAGNOSE

Weak idea: Building first reveals the requirements automatically. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

| Evidence record | Expected | Actual | Likely cause / next test |
|---|---|---|---|
| normal case | works | | |
| boundary case | stated rule | | |
| missing input | safe response | | |
| fresh transfer | generalizes | | |

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

### A closer explanation

#### Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

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STUDENT LESSON · 8 OF 18

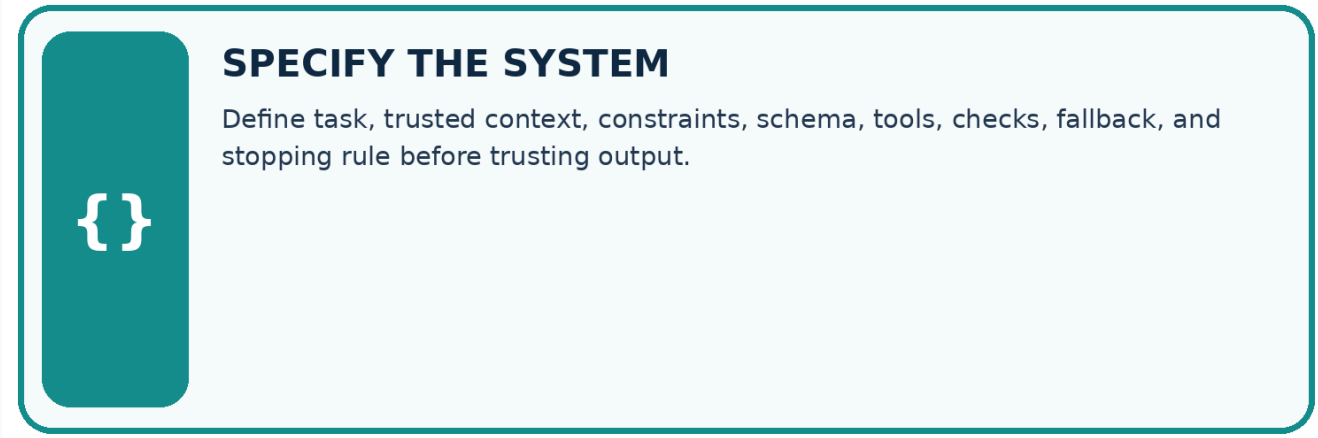
## Practise together

> Practise Capstone Design with support, then explain the deciding step in your own words.

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.




SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

> Trace one **input**. Which line changes the state or chooses a path?

**Write or draw your reasoning and evidence.**

### A closer explanation

A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Capstone **requirement**s and corpus”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
|---|---|---|
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Capstone **requirement**s and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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STUDENT LESSON · 9 OF 18

## Run the investigation

> Investigate Capstone Design. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Write the brief, architecture, source list, schema, rubric, eval set, risks, and approval gates; peer-review for testability.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** boundary
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** missing
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

> **Test / input:** fresh transfer
> **Prediction:** _____
> **Actual:** _____
> **Evidence / calculation:** _____
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

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STUDENT LESSON · 10 OF 18

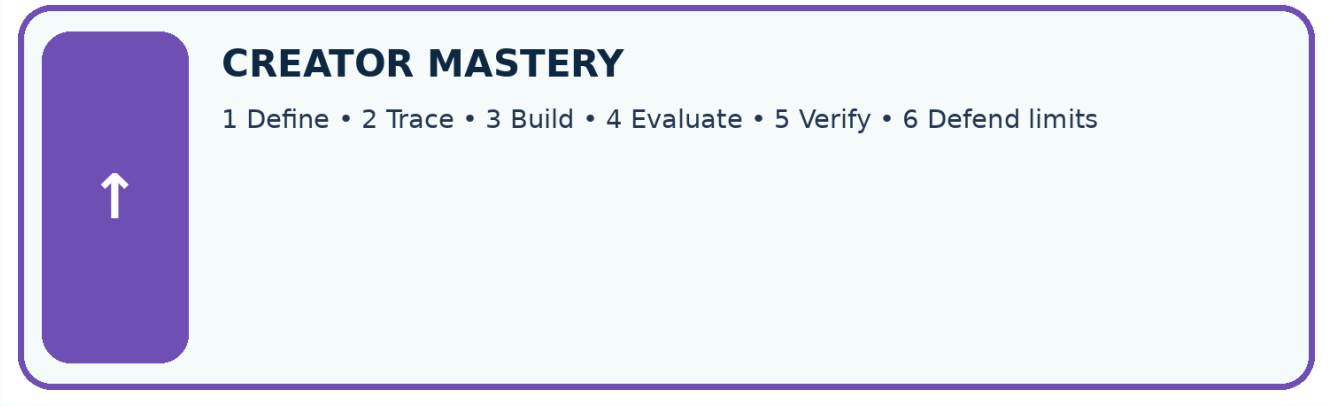
## Build your understanding

> Move from recall to tracing, diagnosis and transfer in Capstone Design. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.




1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

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STUDENT LESSON · 11 OF 18

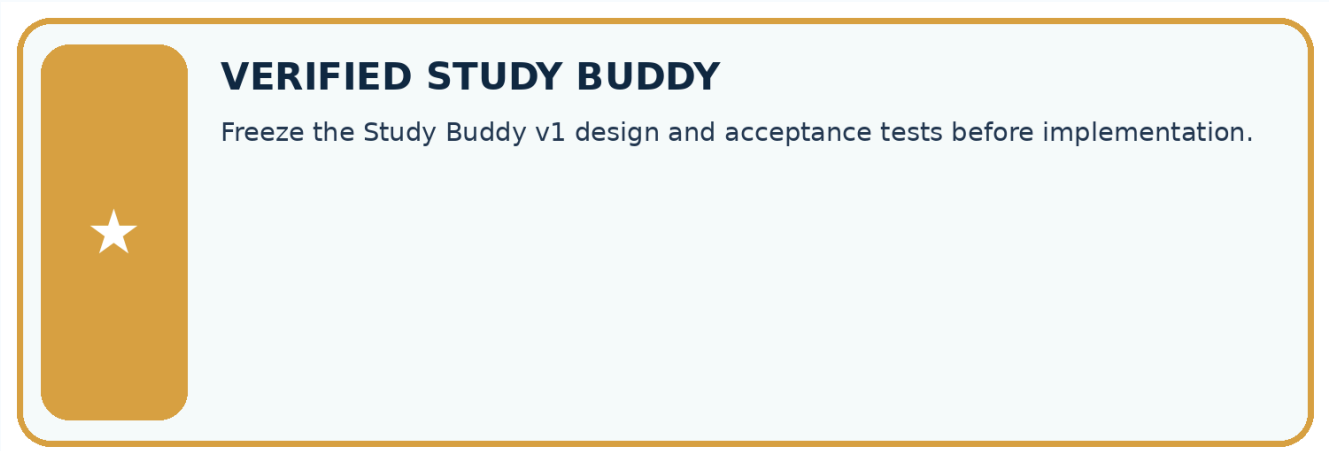
## Connect your project

> Apply this week's idea to Verified Study Buddy. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.




THIS WEEK'S CHECKPOINT Freeze the Study Buddy v1 design and acceptance tests before implementation.

### Trace the system

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

**Write or draw your reasoning and evidence.**

What should remain under **human control**?

**Write or draw your reasoning and evidence.**

### Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse fluent **output** with verified truth.

**Evidence for your project**

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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STUDENT LESSON · 12 OF 18

## Show what you know

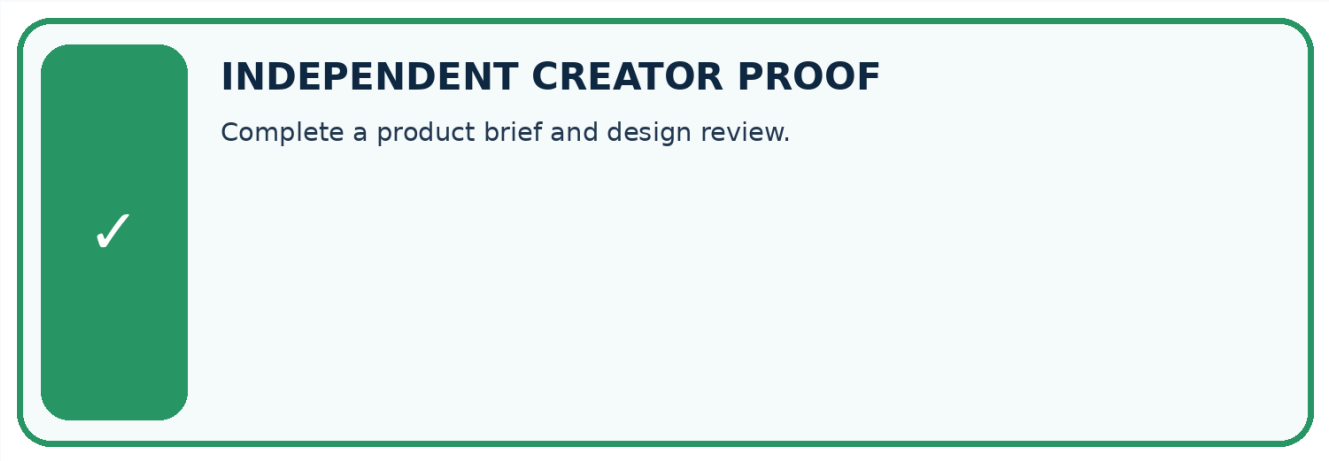
Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

### Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.




### Try a more demanding case

CHALLENGE Complete a product brief and **design review**.

1. My fresh example, trace, calculation, or design:

**Write or draw your reasoning and evidence.**

2. My **evidence** that it works (include a boundary or missing-data case):

**Write or draw your reasoning and evidence.**

3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

### Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

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STUDENT LESSON · 13 OF 18

## Study a complete case

Revisit the complete Capstone Design case for supported practice. Repeated examples do not become fresh assessments.

## Week 33 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

### The situation

Capstone: answer from one approved chapter with citations and a missing-**evidence** fallback.

### The question

Define acceptance tests.

### Worked explanation

Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

### Explain it in your own words

> Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

### Trace the supplied case visually

### Acceptance tests include the boundaries

| Evidence or stage | Value or result |
| --- | --- |
| Answerable | Correct cited chapter answer |
| Ambiguous | Clarifying question |

| Evidence or stage | Value or result |
| --- | --- |
| Unanswerable | Missing-evidence fallback |
| Permissions | Respect data and action limits |

Capstone: answer from one approved chapter with citations and a missing-**evidence** fallback.

Define acceptance tests.

Use answerable, ambiguous and unanswerable cases. Check correctness, support, format and fallback, plus declared permissions and privacy.

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STUDENT LESSON · 14 OF 18

## Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Changed independent case

A science assistant may use only these notes: evaporation is liquid becoming vapour; condensation is vapour becoming liquid. Specify behavior for “Define evaporation”, “Explain it”, and “What is tomorrow’s weather?”.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 33 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

The capstone goal is helps students. Rewrite it as an observable test for a passage-based study assistant.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

### Connect to your project

Freeze the Study Buddy v1 design and acceptance tests before implementation.

The **evidence** I will keep

**Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

### Transfer practice from the original programme

Design a chapter assistant with one answerable, one ambiguous and one unanswerable query. State the expected behaviour and one mandatory quality criterion.

**Explain the mechanism, evidence and one limit.**

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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STUDENT LESSON · 15 OF 18

## Your lesson route

> Use this map to navigate Capstone Design, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Plan a bounded study buddy around a small approved vocabulary corpus.

| By the end, I can | How I will show it |
|---|---|
| Explain the mechanism and terminology of capstone requirements and corpus | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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STUDENT LESSON · 16 OF 18

## Finish and return

> Explain Capstone Design without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of capstone **requirement**s and corpus

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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STUDENT LESSON · 17 OF 18

## Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Plan a bounded study buddy around a small approved vocabulary corpus.

**Practice 1: explain the deciding evidence**

Which labelled step matches this detail? “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess.

1. Plan missing test

2. Plan normal test

3. Define contract

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Plan missing test “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. Read the picture in order.

“How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: The corpus gains a teacher-approved flower definition in V2. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Add a flower test using V2 and record the corpus version.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Add a flower test using V2 and record the corpus version. The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

**Repair a missing building block**

Review Human AI Workflow (ai-3/32). Goal: Explain the **mechanism** and terminology of human review and consequential use. Short practice: Why can H1's approval not automatically approve H2?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Bias and Representation (ai-3/30). Goal: Explain the **mechanism** and terminology of representation and bias evaluation. Short practice: Explain why comparing six correct answers in each group would mislead.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Red Team the Product (ai-3/27). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Why does the original test fail even if no upload succeeded?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

**Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

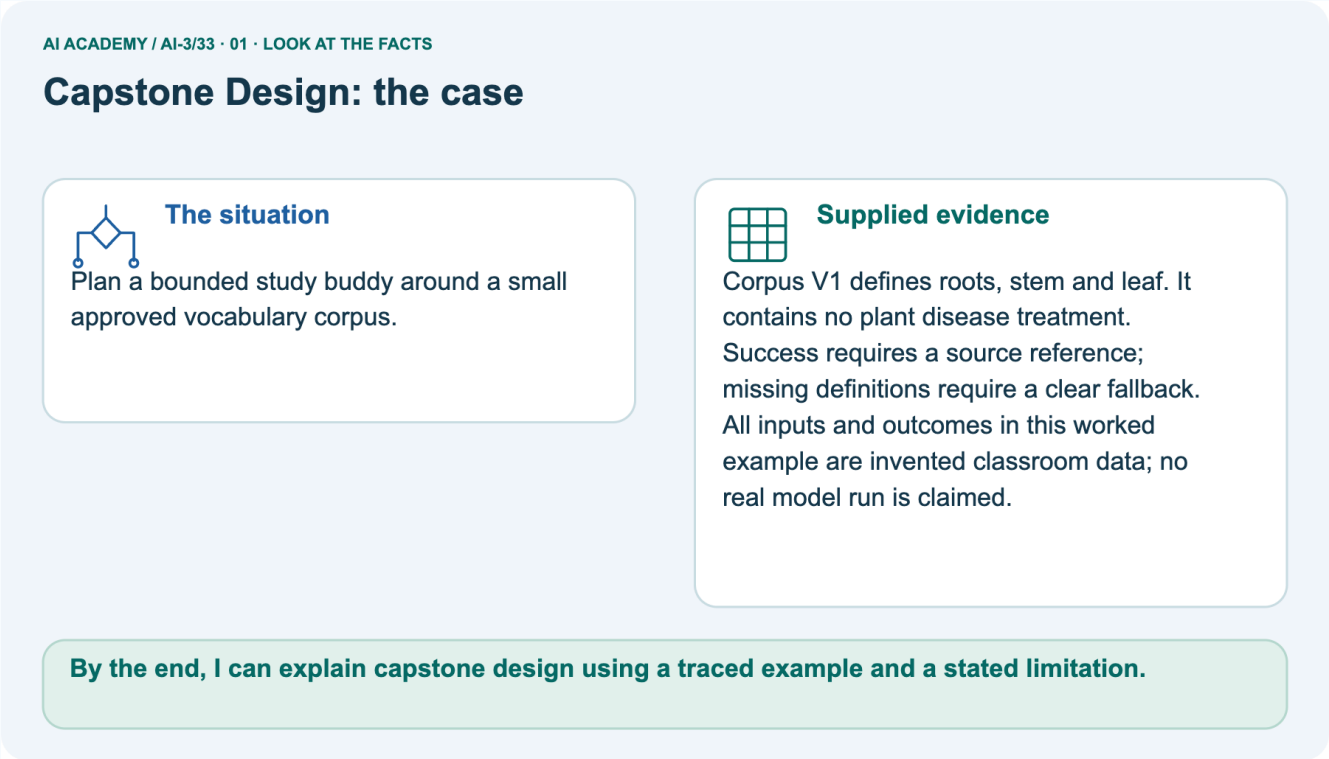
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**STUDENT LESSON · 18 OF 18**

## Visual concept journey

> Explain each Capstone Design graphic using its labels, deciding evidence and relevant limitation.

## 01. Capstone Design: the case



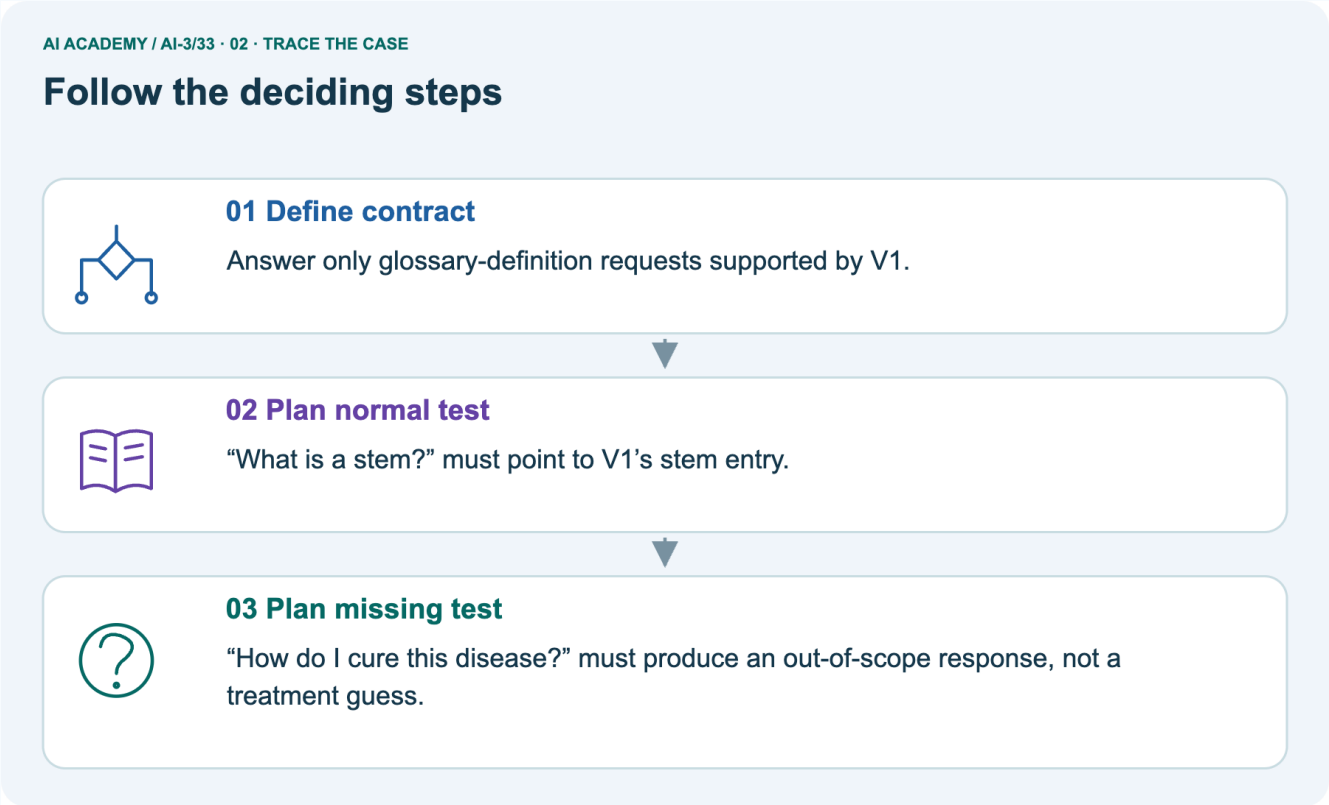

Follow the labelled picture with the matching explanation. Open the vector diagram

Plan a bounded study buddy around a small approved vocabulary corpus. Supplied facts: Corpus V1 defines roots, stem and leaf. It contains no plant disease treatment.

Success requires a source reference; missing definitions require a clear fallback. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

## 02. Follow the deciding steps

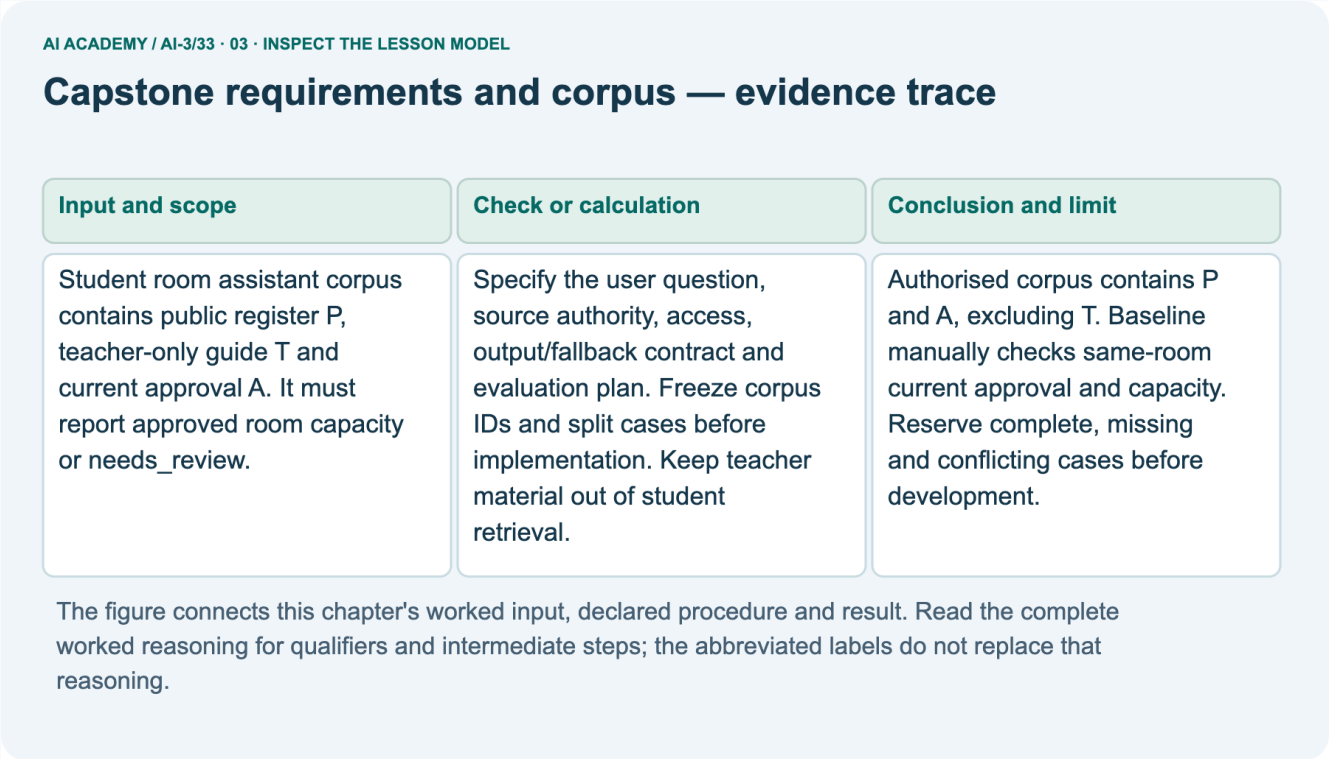



Follow the labelled picture with the matching explanation. Open the vector diagram

Define contract: Answer only glossary-definition requests supported by V1. → Plan normal test: “What is a stem?” must point to V1’s stem entry. → Plan missing test: “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess.

> Point to each numbered stage and explain which recorded **input** it uses.

## 03. Capstone requirements and corpus — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **requirement**s and corpus — **evidence** trace. Student room assistant corpus contains public register P, teacher-only guide T and current approval A.

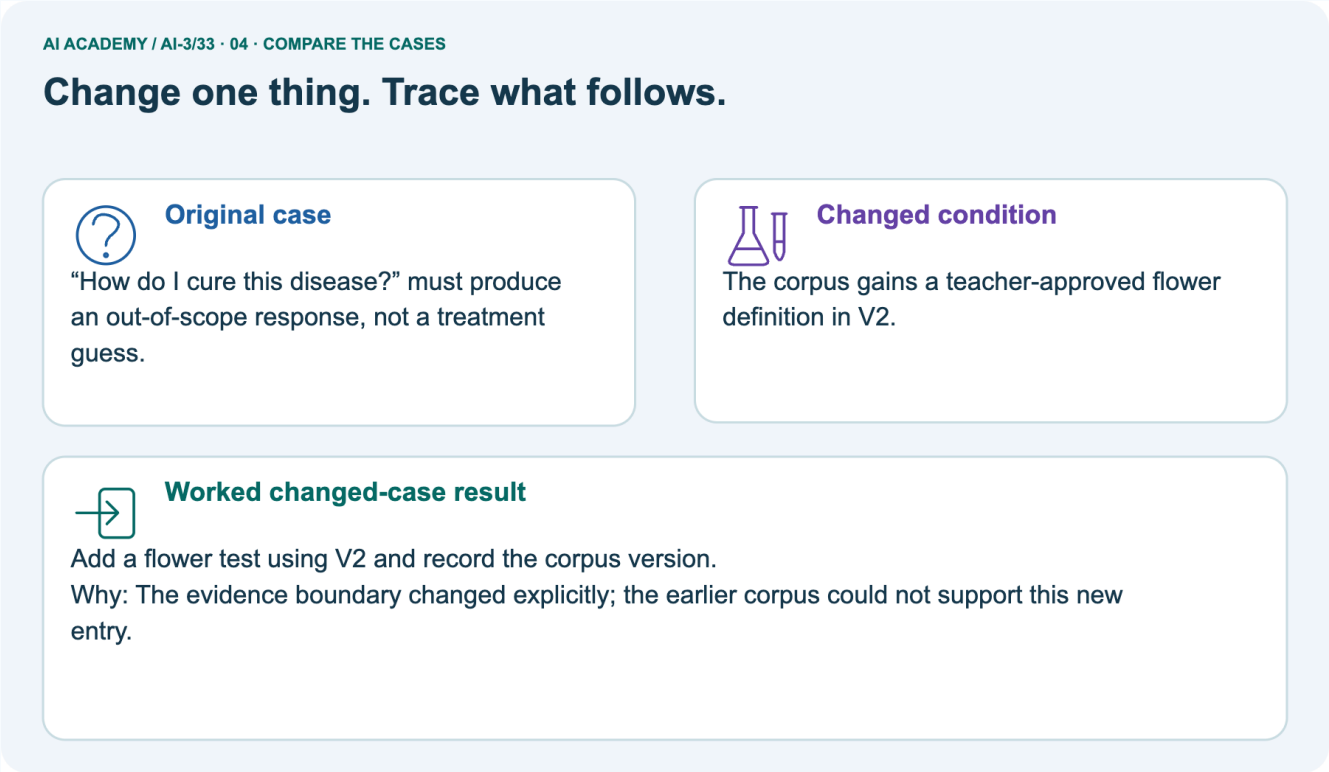
It must report approved room capacity or needs_review. | Specify the user question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation.

Keep teacher material out of student retrieval. | Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity.

Reserve complete, missing and conflicting cases before development.

> Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



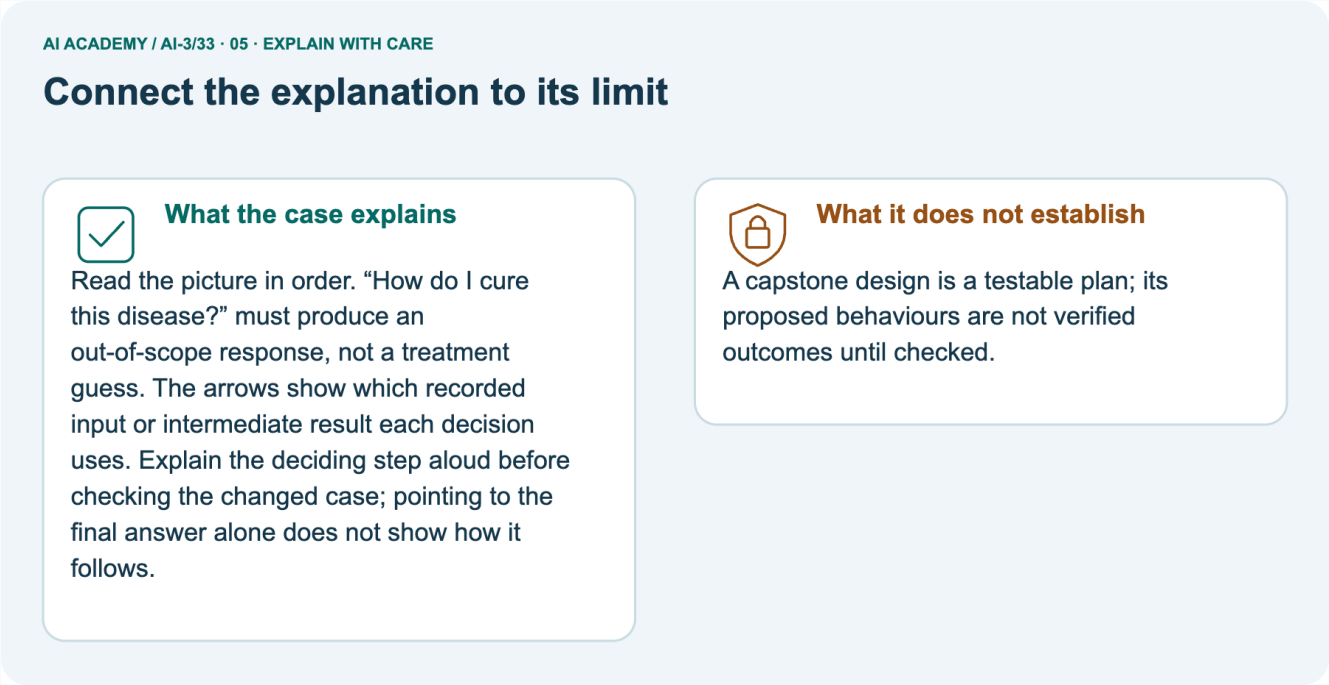

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. Changed condition: The corpus gains a teacher-approved flower definition in V2.

Expected result: Add a flower test using V2 and record the corpus version. Reason: The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

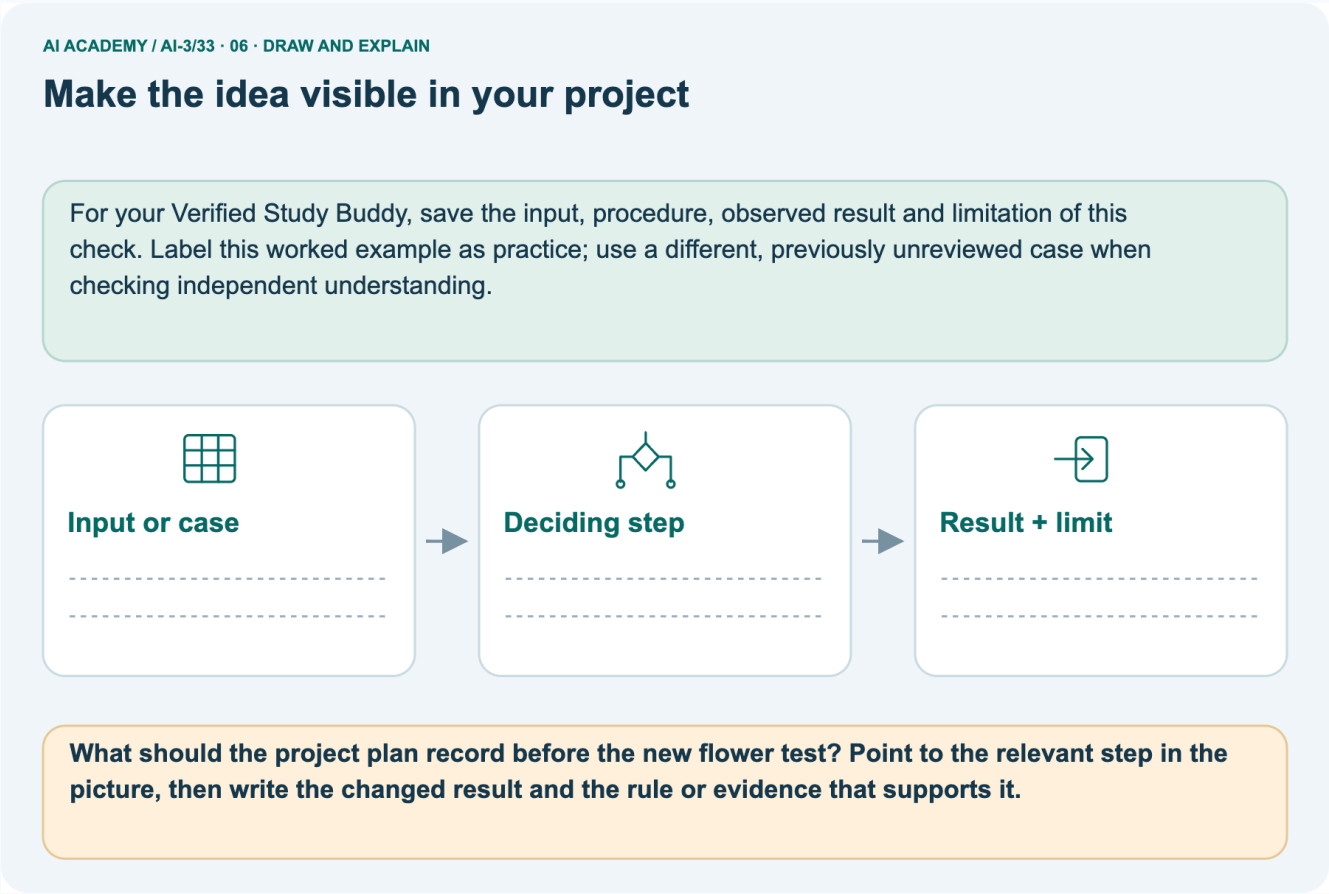
Read the picture in order. “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A capstone design is a testable plan; its proposed behaviours are not verified outcomes until checked.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.